

A proof of Conjecture 00000001016

There is no binary self-dual $[84, 12]$ code

1 The conjecture

Conjecture 00000001016 reads:

Definition: The $[84, 12, 12]$ binary extremal self-dual code is a candidate at the Griesmer boundary. *Conjecture:* This code does not exist; moreover, the automorphism group of every $[84, 12]$ binary self-dual code contains no element of order 11, and the minimum distance is at most 11.

The Chinese text agrees. The claims are: (1) there is no binary self-dual $[84, 12, 12]$ code (in particular no extremal one); (2) no binary self-dual $[84, 12]$ code has an automorphism of order 11; (3) every binary self-dual $[84, 12]$ code has minimum distance ≤ 11 . The statement asserts the existence of nothing: the definition calls the $[84, 12, 12]$ code a “candidate”, and the conjecture asserts that it does not exist. We prove all three claims: (1) is a genuine non-existence theorem, and (2), (3) then hold vacuously.

2 Definitions (standard)

All notions are the textbook ones (e.g. https://en.wikipedia.org/wiki/Dual_code: “A self-dual code is one which is its own dual. This implies that n is even and $\dim C = n/2$.”).

- A binary linear $[n, k]$ code is a k -dimensional subspace $C \subseteq \mathbb{F}_2^n$ (Lean `BinaryCode n`, a `Submodule (ZMod 2) (Fin n -> ZMod 2)`, with `finrank = k`).
- $C^\perp = \{x : x \cdot c = 0 \ \forall c \in C\}$ for the standard dot product $x \cdot y = \sum_i x_i y_i$ (Lean `dual`, the orthogonal complement for the bilinear form `dotForm`; `mem_dual`).
- C is self-dual if $C = C^\perp$ (Lean `IsSelfDual`).
- The minimum distance is the least Hamming weight of a nonzero codeword (Lean `minDistance`, using Mathlib’s `hammingNorm`); an $[n, k, d]$ code has minimum distance d .
- An automorphism of C is a coordinate permutation σ with $c \circ \sigma \in C \iff c \in C$ (Lean `IsAutomorphism`); over $\mathbb{F}_2$ monomial automorphisms are exactly permutations. “The automorphism group contains an element of order 11” means some automorphism σ has `orderOf` $\sigma = 11$.

The word “extremal” needs no definition: claim (1) is proved for every self-dual $[84, 12, 12]$ code.

3 The proof

Lemma 1 (Lean `dotForm_nondegenerate`, `finrank_add_finrank_dual`). *The dot product on $\mathbb{F}_2^n$ is nondegenerate, and $\dim C + \dim C^\perp = n$ for every subspace C .*

Proof. $x \cdot e_i = x_i$, so $x \cdot y = 0$ for all y forces $x = 0$ (and symmetrically). For a nondegenerate bilinear form on a finite-dimensional space, $\dim C^\perp = n - \dim C$ (Mathlib `LinearMap.BilinForm.firank_orthogonal`). $\square$

Theorem 2 (Lean `two_mul_firank_of_selfDual, no_selfDual_84_12`). *A binary self-dual code of length n has $2 \dim C = n$. In particular no binary self-dual code of length 84 has dimension 12 (its dimension is 42).*

Proof. Substitute $C^\perp = C$ in the lemma: $2 \dim C = n$. For $n = 84$, $\dim C = 42 \neq 12$. $\square$

Corollary 3 (Lean `conjecture_1016`). *Claims (1), (2), (3) hold.*

Proof. (1) A self-dual $[84, 12, 12]$ code would be a self-dual code of length 84 and dimension 12. (2), (3): there is no self-dual $[84, 12]$ code, so the universal statements hold vacuously. $\square$

4 Remarks on the reading

The non-existence in (1) is the substance of the result; (2) and (3) are vacuous because the class is empty. Under the nonstandard reading “self-dual = self-orthogonal” ($C \subseteq C^\perp$) the answer changes: the triple extended Golay code $\{(c, c, c) : c \in G_{24}\}$ padded by 12 zero coordinates is a self-orthogonal $[84, 12, 24]$ code, so claim (3) would fail. We use the standard definition $C = C^\perp$. The phrase “at the Griesmer boundary” in the definition is not part of the conjecture and is not used (the Griesmer bound for $k = d = 12$ is $n \geq 31$).

5 Lean formalization

File `lean/Conjecture1016/Basic.lean` (107 lines; only import `Mathlib`). Main theorem `C1016.conjecture_1016`: $(\neg \exists C, \text{IsSelfDual } C \wedge \dim C = 12 \wedge \text{minDistance } C = 12) \wedge (\forall C, \text{IsSelfDual } C \rightarrow \dim C = 12 \rightarrow \forall \sigma, \text{IsAutomorphism } C \sigma \rightarrow \text{orderOf } \sigma \neq 11) \wedge (\forall C, \text{IsSelfDual } C \rightarrow \dim C = 12 \rightarrow \text{minDistance } C \leq 11)$, with C ranging over `BinaryCode 84`. `lake build` succeeds; `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`. No sorry, no `native_decide`, no new axioms.